

MATH1560J Assignment 2 - Sample Answer

Exercise 2.1 (60 pts)

Calculate the first and second derivatives of the following functions.

(i) $f(x) = \sin x + \frac{1}{2} \cot x$ (10 pts)

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\sin x + \frac{1}{2} \cot x \right) \\ &= \cos x - \frac{1}{2} \csc^2 x \\ f''(x) &= \frac{d}{dx} \left(\cos x - \frac{1}{2} \csc^2 x \right) \\ &= -\sin x - \frac{1}{2} \cdot 2 \csc x \cdot \frac{d}{dx} (\csc x) \\ &= -\sin x - \csc x (-\csc x \cot x) \\ &= -\sin x + \csc^2 x \cot x \end{aligned}$$

(ii) $f(x) = 2 \sec x - \csc x$ (10 pts)

$$\begin{aligned} f'(x) &= \frac{d}{dx} (2 \sec x - \csc x) \\ &= 2(\sec x \tan x) - (-\csc x \cot x) \\ &= 2 \sec x \tan x + \csc x \cot x \\ f''(x) &= \frac{d}{dx} (2 \sec x \tan x + \csc x \cot x) \\ &= 2 \left[\frac{d}{dx} (\sec x) \tan x + \sec x \frac{d}{dx} (\tan x) \right] + \left[\frac{d}{dx} (\csc x) \cot x + \csc x \frac{d}{dx} (\cot x) \right] \\ &= 2 \left[(\sec x \tan x) \tan x + \sec x (\sec^2 x) \right] + \left[(-\csc x \cot x) \cot x + \csc x (-\csc^2 x) \right] \\ &= 2 \sec x \tan^2 x + 2 \sec^3 x - \csc x \cot^2 x - \csc^3 x \end{aligned}$$

(iii) $f(x) = \frac{x}{2-\tan x}$ (10 pts)

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} \left(\frac{x}{2-\tan x} \right) \\
 &= \frac{\frac{d}{dx}(x)(2-\tan x) - x \frac{d}{dx}(2-\tan x)}{(2-\tan x)^2} \\
 &= \frac{1 \cdot (2-\tan x) - x(0-\sec^2 x)}{(2-\tan x)^2} \\
 &= \frac{2-\tan x + x \sec^2 x}{(2-\tan x)^2} \\
 f''(x) &= \frac{d}{dx} \left(\frac{2-\tan x + x \sec^2 x}{(2-\tan x)^2} \right) \\
 &= \frac{\frac{d}{dx}(2-\tan x + x \sec^2 x)(2-\tan x)^2 - (2-\tan x + x \sec^2 x) \frac{d}{dx}((2-\tan x)^2)}{(2-\tan x)^4}
 \end{aligned}$$

$$\begin{aligned}
 \text{where } \frac{d}{dx}(2-\tan x + x \sec^2 x) &= -\sec^2 x + (1 \cdot \sec^2 x + x \cdot 2 \sec x (\sec x \tan x)) \\
 &= -\sec^2 x + \sec^2 x + 2x \sec^2 x \tan x = 2x \sec^2 x \tan x
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \frac{d}{dx}((2-\tan x)^2) &= 2(2-\tan x) \frac{d}{dx}(2-\tan x) \\
 &= 2(2-\tan x)(-\sec^2 x) = -2 \sec^2 x (2-\tan x)
 \end{aligned}$$

$$\begin{aligned}
 f''(x) &= \frac{(2x \sec^2 x \tan x)(2-\tan x)^2 - (2-\tan x + x \sec^2 x)(-2 \sec^2 x (2-\tan x))}{(2-\tan x)^4} \\
 &= \frac{(2x \sec^2 x \tan x)(2-\tan x) + 2 \sec^2 x (2-\tan x + x \sec^2 x)}{(2-\tan x)^3} \\
 &= \frac{2 \sec^2 x [x \tan x (2-\tan x) + (2-\tan x + x \sec^2 x)]}{(2-\tan x)^3} \\
 &= \frac{2 \sec^2 x [2x \tan x - x \tan^2 x + 2 - \tan x + x \sec^2 x]}{(2-\tan x)^3} \\
 &= \frac{2 \sec^2 x [2x \tan x + x + 2 - \tan x]}{(2-\tan x)^3}
 \end{aligned}$$

(iv) $f(x) = \frac{\sec x}{1+\sec x}$ (10 pts)

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} \left(\frac{\sec x}{1 + \sec x} \right) \\
 &= \frac{\frac{d}{dx} (\sec x) (1 + \sec x) - \sec x \frac{d}{dx} (1 + \sec x)}{(1 + \sec x)^2} \\
 &= \frac{(\sec x \tan x)(1 + \sec x) - \sec x(\sec x \tan x)}{(1 + \sec x)^2} \\
 &= \frac{\sec x \tan x + \sec^2 x \tan x - \sec^2 x \tan x}{(1 + \sec x)^2} \\
 &= \frac{\sec x \tan x}{(1 + \sec x)^2}
 \end{aligned}$$

$$\begin{aligned}
 f''(x) &= \frac{d}{dx} \left(\frac{\sec x \tan x}{(1 + \sec x)^2} \right) \\
 &= \frac{\frac{d}{dx} (\sec x \tan x) (1 + \sec x)^2 - (\sec x \tan x) \frac{d}{dx} ((1 + \sec x)^2)}{(1 + \sec x)^4}
 \end{aligned}$$

where $\frac{d}{dx} (\sec x \tan x) = (\sec x \tan x) \tan x + \sec x (\sec^2 x) = \sec x \tan^2 x + \sec^3 x$

and $\frac{d}{dx} ((1 + \sec x)^2) = 2(1 + \sec x) \frac{d}{dx} (1 + \sec x) = 2(1 + \sec x)(\sec x \tan x)$

$$\begin{aligned}
 f''(x) &= \frac{(\sec x \tan^2 x + \sec^3 x)(1 + \sec x)^2 - (\sec x \tan x)(2(1 + \sec x) \sec x \tan x)}{(1 + \sec x)^4} \\
 &= \frac{(\sec x \tan^2 x + \sec^3 x)(1 + \sec x) - 2 \sec^2 x \tan^2 x}{(1 + \sec x)^3} \\
 &= \frac{\sec x \tan^2 x + \sec^2 x \tan^2 x + \sec^3 x + \sec^4 x - 2 \sec^2 x \tan^2 x}{(1 + \sec x)^3} \\
 &= \frac{\sec^4 x + \sec^3 x + \sec x \tan^2 x - \sec^2 x \tan^2 x}{(1 + \sec x)^3}
 \end{aligned}$$

(v) $f(x) = \frac{1-\sec x}{\tan x}$ (10 pts)

We can simplify $f(x)$ first:

$$f(x) = \frac{1 - \frac{1}{\cos x}}{\frac{\sin x}{\cos x}} = \frac{\frac{\cos x - 1}{\cos x}}{\frac{\sin x}{\cos x}} = \frac{\cos x - 1}{\sin x} = \frac{\cos x}{\sin x} - \frac{1}{\sin x} = \cot x - \csc x$$

$$\begin{aligned} f'(x) &= \frac{d}{dx} (\cot x - \csc x) \\ &= -\csc^2 x - (-\csc x \cot x) \\ &= -\csc^2 x + \csc x \cot x \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{d}{dx} (-\csc^2 x + \csc x \cot x) \\ &= -2 \csc x \frac{d}{dx} (\csc x) + \left[\frac{d}{dx} (\csc x) \cot x + \csc x \frac{d}{dx} (\cot x) \right] \\ &= -2 \csc x (-\csc x \cot x) + [(-\csc x \cot x) \cot x + \csc x (-\csc^2 x)] \\ &= 2 \csc^2 x \cot x - \csc x \cot^2 x - \csc^3 x \end{aligned}$$

(vi) $f(x) = x^2 \sin x \tan x$ (10 pts)

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} (x^2) \sin x \tan x + x^2 \frac{d}{dx} (\sin x) \tan x + x^2 \sin x \frac{d}{dx} (\tan x) \\
 &= 2x \sin x \tan x + x^2 \sin x + x^2 \sin x \sec^2 x \\
 f''(x) &= \frac{d}{dx} (2x \sin x \tan x) + \frac{d}{dx} (x^2 \sin x) + \frac{d}{dx} (x^2 \sin x \sec^2 x) \\
 \frac{d}{dx} (2x \sin x \tan x) &= 2 \left[\frac{d}{dx} (x) \sin x \tan x + x \frac{d}{dx} (\sin x) \tan x + x \sin x \frac{d}{dx} (\tan x) \right] \\
 &= 2 [1 \cdot \sin x \tan x + x \cos x \tan x + x \sin x \sec^2 x] \\
 &= 2 \sin x \tan x + 2x \sin x + 2x \sin x \sec^2 x \\
 \frac{d}{dx} (x^2 \sin x) &= \frac{d}{dx} (x^2) \sin x + x^2 \frac{d}{dx} (\sin x) \\
 &= 2x \sin x + x^2 \cos x \\
 \frac{d}{dx} (x^2 \sin x \sec^2 x) &= \frac{d}{dx} (x^2) \sin x \sec^2 x + x^2 \frac{d}{dx} (\sin x) \sec^2 x + x^2 \sin x \frac{d}{dx} (\sec^2 x) \\
 &= 2x \sin x \sec^2 x + x^2 \cos x \sec^2 x + x^2 \sin x (2 \sec x \cdot \sec x \tan x) \\
 &= 2x \sin x \sec^2 x + x^2 \sec x + 2x^2 \sin x \sec^2 x \tan x \\
 f''(x) &= (2 \sin x \tan x + 2x \sin x + 2x \sin x \sec^2 x) \\
 &\quad + (2x \sin x + x^2 \cos x) \\
 &\quad + (2x \sin x \sec^2 x + x^2 \sec x + 2x^2 \sin x \sec^2 x \tan x) \\
 &= 2 \sin x \tan x + 4x \sin x + 4x \sin x \sec^2 x \\
 &\quad + x^2 \sec x + x^2 \cos x + 2x^2 \sin x \sec^2 x \tan x
 \end{aligned}$$

Exercise 2.2 (10 pts)

Find constants A and B such that $y = A \sin x + B \cos x$ satisfies $y'' + y' - 2y = \sin x$.

$$\begin{aligned}
 y &= A \sin x + B \cos x \\
 y' &= A \cos x - B \sin x \\
 y'' &= -A \sin x - B \cos x
 \end{aligned}$$

Substitute into the differential equation $y'' + y' - 2y = \sin x$:

$$\begin{aligned}
 (-A \sin x - B \cos x) + (A \cos x - B \sin x) - 2(A \sin x + B \cos x) &= \sin x \\
 (-A - B - 2A) \sin x + (-B + A - 2B) \cos x &= \sin x \\
 (-3A - B) \sin x + (A - 3B) \cos x &= 1 \cdot \sin x + 0 \cdot \cos x
 \end{aligned}$$

Equating the coefficients of $\sin x$ and $\cos x$:

$$-3A - B = 1 \quad (1)$$

$$A - 3B = 0 \quad (2)$$

From (2), $A = 3B$. Substitute into (1):

$$-3(3B) - B = 1$$

$$-9B - B = 1$$

$$-10B = 1$$

$$B = -\frac{1}{10}$$

Then $A = 3B = 3\left(-\frac{1}{10}\right) = -\frac{3}{10}$. The constants are $A = -\frac{3}{10}$ and $B = -\frac{1}{10}$.

Exercise 2.3 (20 pts)

The curve $y = [x]/\sqrt{2-x^2}$ is called a bullet-nose curve.

(i) Find an equation of the tangent line to this curve at the point $(1, 1)$. (10 pts)

The equation of the curve is $y^2(2-x^2) = x^2$. The text uses $y = [x]/\sqrt{2-x^2}$ which includes the floor function $[x]$. Since the point $(1, 1)$ is given, and $x = 1$ is an integer, we assume $y = x/\sqrt{2-x^2}$ is intended near $x = 1$ for differentiability, or the problem means to use the algebraic form $y^2(2-x^2) = x^2$ for the continuous part of the curve near $(1, 1)$. Using the latter:

$$y^2(2-x^2) = x^2$$

Differentiate implicitly with respect to x :

$$\frac{d}{dx}(y^2(2-x^2)) = \frac{d}{dx}(x^2)$$

$$2y\frac{dy}{dx}(2-x^2) + y^2(-2x) = 2x$$

$$2y(2-x^2)\frac{dy}{dx} = 2x + 2xy^2$$

$$\frac{dy}{dx} = \frac{2x(1+y^2)}{2y(2-x^2)} = \frac{x(1+y^2)}{y(2-x^2)}$$

At the point $(1, 1)$:

$$m = \left. \frac{dy}{dx} \right|_{(1,1)} = \frac{1(1+1^2)}{1(2-1^2)} = \frac{2}{1} = 2$$

The equation of the tangent line (point-slope form):

$$y - 1 = 2(x - 1) \implies y = 2x - 2 + 1 \implies y = 2x - 1$$

(ii) Illustrate part (i) by sketch the curve and the tangent line on the same coordinate system. (10 pts)

This is a sketching problem which is best solved graphically. The curve $y^2(2-x^2) = x^2$ is an algebraic curve. The tangent line is $y = 2x - 1$.

Exercise 2.4 (10 pts)

If $y = f(u)$ and $u = g(x)$ where f and g are twice differentiable functions, show that

$$\frac{d^2y}{dx^2} = \frac{d^2y}{du^2} \left(\frac{du}{dx} \right)^2 + \frac{dy}{du} \frac{d^2u}{dx^2}$$

By the Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Now, differentiate $\frac{dy}{dx}$ with respect to x using the Product Rule and the Chain Rule:

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\left(\frac{dy}{du} \frac{du}{dx} \right) \right) \\ &= \frac{d}{dx} \left(\left(\frac{dy}{du} \right) \right) \frac{du}{dx} + \frac{dy}{du} \frac{d}{dx} \left(\left(\frac{du}{dx} \right) \right) \\ &= \left(\frac{d}{du} \left(\frac{dy}{du} \right) \frac{du}{dx} \right) \frac{du}{dx} + \frac{dy}{du} \frac{d^2u}{dx^2} \\ &= \frac{d^2y}{du^2} \left(\frac{du}{dx} \right) \frac{du}{dx} + \frac{dy}{du} \frac{d^2u}{dx^2} \\ &= \frac{d^2y}{du^2} \left(\frac{du}{dx} \right)^2 + \frac{dy}{du} \frac{d^2u}{dx^2} \end{aligned}$$

Exercise 2.5 (40 pts)

Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

(i) (cardioid) $x^2 + y^2 = (2x^2 + 2y^2 - x)^2$ at $(0, 1/2)$. (10 pts)

Differentiate implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^2 + y^2) &= \frac{d}{dx}((2x^2 + 2y^2 - x)^2) \\ 2x + 2y \frac{dy}{dx} &= 2(2x^2 + 2y^2 - x) \cdot \frac{d}{dx}(2x^2 + 2y^2 - x) \\ 2x + 2y \frac{dy}{dx} &= 2(2x^2 + 2y^2 - x) \cdot (4x + 4y \frac{dy}{dx} - 1)\end{aligned}$$

At the point $(0, 1/2)$, we have $x = 0$ and $y = 1/2$. Substitute these values:

$$\begin{aligned}2(0) + 2\left(\frac{1}{2}\right) \frac{dy}{dx} \Big|_{(0,1/2)} &= 2\left(2(0)^2 + 2\left(\frac{1}{2}\right)^2 - 0\right) \cdot \left(4(0) + 4\left(\frac{1}{2}\right) \frac{dy}{dx} \Big|_{(0,1/2)} - 1\right) \\ \frac{dy}{dx} \Big|_{(0,1/2)} &= 2\left(\frac{1}{2}\right) \cdot \left(2 \frac{dy}{dx} \Big|_{(0,1/2)} - 1\right) \\ \frac{dy}{dx} \Big|_{(0,1/2)} &= 2 \frac{dy}{dx} \Big|_{(0,1/2)} - 1 \\ 1 &= \frac{dy}{dx} \Big|_{(0,1/2)}\end{aligned}$$

The slope is $m = 1$. The tangent line is:

$$y - \frac{1}{2} = 1(x - 0) \implies y = x + \frac{1}{2}$$

(ii) (astroid) $x^{2/3} + y^{2/3} = 4$ at $(-3\sqrt{3}, 1)$. (10 pts)

Differentiate implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^{2/3} + y^{2/3}) &= \frac{d}{dx}(4) \\ \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3} \frac{dy}{dx} &= 0 \\ y^{-1/3} \frac{dy}{dx} &= -x^{-1/3} \\ \frac{dy}{dx} &= -\frac{x^{-1/3}}{y^{-1/3}} = -\left(\frac{y}{x}\right)^{1/3}\end{aligned}$$

At the point $(-3\sqrt{3}, 1)$:

$$m = \frac{dy}{dx} \Big|_{(-3\sqrt{3}, 1)} = -\left(\frac{1}{-3\sqrt{3}}\right)^{1/3} = -\left(-\frac{1}{3\sqrt{3}}\right)^{1/3} = -\left(-\left(\frac{1}{3^{3/2}}\right)^{1/3}\right) = -\left(-\frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}}$$

The tangent line is:

$$y - 1 = \frac{1}{\sqrt{3}}(x - (-3\sqrt{3})) \implies y - 1 = \frac{1}{\sqrt{3}}x + 3 \implies y = \frac{\sqrt{3}}{3}x + 4$$

(iii) (lemniscate) $2(x^2 + y^2)^2 = 25(x^2 - y^2)$ at $(3, 1)$. (10 pts)

Differentiate implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(2(x^2 + y^2)^2) &= \frac{d}{dx}(25(x^2 - y^2)) \\ 4(x^2 + y^2) \cdot \frac{d}{dx}(x^2 + y^2) &= 25(2x - 2y\frac{dy}{dx}) \\ 4(x^2 + y^2)(2x + 2y\frac{dy}{dx}) &= 50x - 50y\frac{dy}{dx} \\ 8(x^2 + y^2)x + 8(x^2 + y^2)y\frac{dy}{dx} &= 50x - 50y\frac{dy}{dx} \\ [8y(x^2 + y^2) + 50y]\frac{dy}{dx} &= 50x - 8x(x^2 + y^2) \\ \frac{dy}{dx} &= \frac{50x - 8x(x^2 + y^2)}{50y + 8y(x^2 + y^2)}\end{aligned}$$

At the point $(3, 1)$: $x^2 + y^2 = 3^2 + 1^2 = 10$.

$$m = \left. \frac{dy}{dx} \right|_{(3,1)} = \frac{50(3) - 8(3)(10)}{50(1) + 8(1)(10)} = \frac{150 - 240}{50 + 80} = \frac{-90}{130} = -\frac{9}{13}$$

The tangent line is:

$$y - 1 = -\frac{9}{13}(x - 3) \implies 13(y - 1) = -9(x - 3) \implies 13y - 13 = -9x + 27 \implies 9x + 13y = 40$$

(iv) (devil's curve) $y^2(y^2 - 4) = x^2(x^2 - 5)$ at $(0, -2)$. (10 pts)

The equation is $y^4 - 4y^2 = x^4 - 5x^2$. Differentiate implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(y^4 - 4y^2) &= \frac{d}{dx}(x^4 - 5x^2) \\ 4y^3\frac{dy}{dx} - 8y\frac{dy}{dx} &= 4x^3 - 10x \\ (4y^3 - 8y)\frac{dy}{dx} &= 4x^3 - 10x \\ \frac{dy}{dx} &= \frac{4x^3 - 10x}{4y^3 - 8y}\end{aligned}$$

At the point $(0, -2)$:

$$m = \left. \frac{dy}{dx} \right|_{(0, -2)} = \frac{4(0)^3 - 10(0)}{4(-2)^3 - 8(-2)} = \frac{0}{4(-8) + 16} = \frac{0}{-32 + 16} = 0$$

The slope is $m = 0$. The tangent line is:

$$y - (-2) = 0(x - 0) \implies y + 2 = 0 \implies y = -2$$

Exercise 2.6 (120 pts)

Given the hyperbolic functions $\sinh x = \frac{e^x - e^{-x}}{2}$, $\cosh x = \frac{e^x + e^{-x}}{2}$, $\tanh x = \frac{\sinh x}{\cosh x}$, $\coth x = \frac{\cosh x}{\sinh x}$, $\operatorname{sech} x = \frac{1}{\cosh x}$, $\operatorname{csch} x = \frac{1}{\sinh x}$.

(i) Show the following derivatives:

(a) $\frac{d}{dx}(\sinh x) = \cosh x$ (10 pts)

$$\begin{aligned} \frac{d}{dx}(\sinh x) &= \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) \\ &= \frac{1}{2} \left(\frac{d}{dx}(e^x) - \frac{d}{dx}(e^{-x}) \right) \\ &= \frac{1}{2}(e^x - e^{-x}(-1)) \\ &= \frac{e^x + e^{-x}}{2} = \cosh x \end{aligned}$$

(b) $\frac{d}{dx}(\cosh x) = \sinh x$ (10 pts)

$$\begin{aligned} \frac{d}{dx}(\cosh x) &= \frac{d}{dx} \left(\frac{e^x + e^{-x}}{2} \right) \\ &= \frac{1}{2} \left(\frac{d}{dx}(e^x) + \frac{d}{dx}(e^{-x}) \right) \\ &= \frac{1}{2}(e^x + e^{-x}(-1)) \\ &= \frac{e^x - e^{-x}}{2} = \sinh x \end{aligned}$$

(c) $\frac{d}{dx}(\tanh x) = \text{sech}^2 x$ (10 pts)

$$\begin{aligned}\frac{d}{dx}(\tanh x) &= \frac{d}{dx} \left(\frac{\sinh x}{\cosh x} \right) \\ &= \frac{\frac{d}{dx}(\sinh x) \cosh x - \sinh x \frac{d}{dx}(\cosh x)}{\cosh^2 x} \\ &= \frac{(\cosh x) \cosh x - \sinh x (\sinh x)}{\cosh^2 x} \\ &= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x}\end{aligned}$$

Using the identity $\cosh^2 x - \sinh^2 x = 1$:

$$= \frac{1}{\cosh^2 x} = \left(\frac{1}{\cosh x} \right)^2 = \text{sech}^2 x$$

(d) $\frac{d}{dx}(\text{csch } x) = -\text{csch } x \coth x$ (10 pts)

$$\begin{aligned}\frac{d}{dx}(\text{csch } x) &= \frac{d}{dx} \left(\frac{1}{\sinh x} \right) \\ &= -\frac{1}{\sinh^2 x} \frac{d}{dx}(\sinh x) \\ &= -\frac{1}{\sinh^2 x} (\cosh x) \\ &= -\frac{1}{\sinh x} \cdot \frac{\cosh x}{\sinh x} \\ &= -\text{csch } x \coth x\end{aligned}$$

(e) $\frac{d}{dx}(\text{sech } x) = -\text{sech } x \tanh x$ (10 pts)

$$\begin{aligned}\frac{d}{dx}(\text{sech } x) &= \frac{d}{dx} \left(\frac{1}{\cosh x} \right) \\ &= -\frac{1}{\cosh^2 x} \frac{d}{dx}(\cosh x) \\ &= -\frac{1}{\cosh^2 x} (\sinh x) \\ &= -\frac{1}{\cosh x} \cdot \frac{\sinh x}{\cosh x} \\ &= -\text{sech } x \tanh x\end{aligned}$$

(f) $\frac{d}{dx}(\coth x) = -\text{csch}^2 x$ (10 pts)

$$\begin{aligned}\frac{d}{dx}(\coth x) &= \frac{d}{dx} \left(\frac{\cosh x}{\sinh x} \right) \\ &= \frac{\frac{d}{dx}(\cosh x) \sinh x - \cosh x \frac{d}{dx}(\sinh x)}{\sinh^2 x} \\ &= \frac{(\sinh x) \sinh x - \cosh x (\cosh x)}{\sinh^2 x} \\ &= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x}\end{aligned}$$

Using the identity $\sinh^2 x - \cosh^2 x = -(\cosh^2 x - \sinh^2 x) = -1$:

$$= \frac{-1}{\sinh^2 x} = - \left(\frac{1}{\sinh x} \right)^2 = -\text{csch}^2 x$$

(ii) Show the following derivatives:

(a) $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$ (10 pts)

Let $y = \sinh^{-1} x$. Then $x = \sinh y$. Differentiate implicitly with respect to x :

$$1 = \cosh y \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\cosh y}$$

We use the identity $\cosh^2 y - \sinh^2 y = 1$, so $\cosh y = \sqrt{1 + \sinh^2 y}$ (since $\cosh y > 0$).

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 + \sinh^2 y}} = \frac{1}{\sqrt{1 + x^2}}$$

(b) $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$ (10 pts)

Let $y = \cosh^{-1} x$. Then $x = \cosh y$. Differentiate implicitly with respect to x :

$$1 = \sinh y \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\sinh y}$$

We use the identity $\cosh^2 y - \sinh^2 y = 1$, so $\sinh y = \sqrt{\cosh^2 y - 1}$ (since $x > 1$, $y > 0$, $\sinh y > 0$).

$$\frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2 y - 1}} = \frac{1}{\sqrt{x^2 - 1}}$$

(c) $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$ (10 pts)

Let $y = \tanh^{-1} x$. Then $x = \tanh y$. Differentiate implicitly with respect to x :

$$1 = \operatorname{sech}^2 y \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y} = \cosh^2 y$$

We use the identity $\operatorname{sech}^2 y = 1 - \tanh^2 y$:

$$\frac{dy}{dx} = \frac{1}{1 - \tanh^2 y} = \frac{1}{1 - x^2}$$

(d) $\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{x^2+1}}$ (10 pts)

Let $y = \operatorname{csch}^{-1} x$. Then $x = \operatorname{csch} y$. Differentiate implicitly with respect to x :

$$1 = -\operatorname{csch} y \coth y \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\operatorname{csch} y \coth y}$$

We use the identity $\coth^2 y - \operatorname{csch}^2 y = 1$, so $\coth y = \pm\sqrt{1 + \operatorname{csch}^2 y}$. The sign of $\coth y$ is the same as the sign of y , and the sign of $\operatorname{csch} y = x$ is the same as the sign of y (because $x = 1/\sinh y$, $\sinh y$ has the same sign as y). Thus, $\operatorname{csch} y \coth y = |x|\sqrt{1 + x^2}$.

$$\frac{dy}{dx} = -\frac{1}{|x|\sqrt{1 + x^2}}$$

(e) $\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$ (10 pts)

Let $y = \operatorname{sech}^{-1} x$. Then $x = \operatorname{sech} y$. Differentiate implicitly with respect to x :

$$1 = -\operatorname{sech} y \tanh y \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\operatorname{sech} y \tanh y}$$

We use $\tanh^2 y + \operatorname{sech}^2 y = 1$, so $\tanh y = \sqrt{1 - \operatorname{sech}^2 y}$ (since the domain of $\operatorname{sech}^{-1} x$ is $(0, 1]$, $y \geq 0$, $\tanh y \geq 0$).

$$\frac{dy}{dx} = -\frac{1}{x\sqrt{1 - x^2}}$$

(f) $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$ **(10 pts)**

Let $y = \coth^{-1} x$. Then $x = \coth y$. Differentiate implicitly with respect to x :

$$1 = -\operatorname{csch}^2 y \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\operatorname{csch}^2 y} = -\sinh^2 y$$

We use the identity $\coth^2 y - \operatorname{csch}^2 y = 1$, so $\operatorname{csch}^2 y = \coth^2 y - 1$.

$$\frac{dy}{dx} = -\frac{1}{\coth^2 y - 1} = -\frac{1}{x^2 - 1} = \frac{1}{1 - x^2}$$

Exercise 2.7 (20 pts)

Find the derivative of the following functions using logarithmic differentiation.

(i) $y = (\sin x)^{\ln x}$ **(10 pts)**

Take the natural logarithm of both sides:

$$\ln y = \ln((\sin x)^{\ln x}) = (\ln x) \ln(\sin x)$$

Differentiate implicitly with respect to x :

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \frac{d}{dx} (\ln x) \ln(\sin x) + \ln x \frac{d}{dx} (\ln(\sin x)) \\ &= \frac{1}{x} \ln(\sin x) + \ln x \cdot \frac{1}{\sin x} \cdot \cos x \\ &= \frac{\ln(\sin x)}{x} + \ln x \cot x \\ \frac{dy}{dx} &= y \left(\frac{\ln(\sin x)}{x} + \ln x \cot x \right) \\ &= (\sin x)^{\ln x} \left(\frac{\ln(\sin x)}{x} + \ln x \cot x \right) \end{aligned}$$

(ii) $y = (\tan x)^{1/x}$ **(10 pts)**

Take the natural logarithm of both sides:

$$\ln y = \ln((\tan x)^{1/x}) = \frac{1}{x} \ln(\tan x)$$

Differentiate implicitly with respect to x :

$$\begin{aligned}
 \frac{1}{y} \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{1}{x} \right) \ln(\tan x) + \frac{1}{x} \frac{d}{dx} (\ln(\tan x)) \\
 &= -\frac{1}{x^2} \ln(\tan x) + \frac{1}{x} \cdot \frac{1}{\tan x} \cdot \sec^2 x \\
 &= -\frac{\ln(\tan x)}{x^2} + \frac{1}{x} \cdot \cot x \sec^2 x \\
 &= -\frac{\ln(\tan x)}{x^2} + \frac{1}{x} \cdot \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} \\
 &= -\frac{\ln(\tan x)}{x^2} + \frac{1}{x} \cdot \frac{1}{\sin x \cos x} \\
 \frac{dy}{dx} &= y \left(-\frac{\ln(\tan x)}{x^2} + \frac{1}{x \sin x \cos x} \right) \\
 &= (\tan x)^{1/x} \left(\frac{1}{x \sin x \cos x} - \frac{\ln(\tan x)}{x^2} \right)
 \end{aligned}$$

Exercise 2.8 (30 pts)

Express the limit as a derivative and evaluate.

(i) $\lim_{x \rightarrow 1} \frac{x^{17} - 1}{x - 1}$ (10 pts)

Let $f(x) = x^{17}$. The limit is in the form of the derivative definition $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ with $a = 1$ and $f(1) = 1^{17} = 1$. The limit is $f'(1)$.

$$f'(x) = \frac{d}{dx} (x^{17}) = 17x^{16}$$

$$\lim_{x \rightarrow 1} \frac{x^{17} - 1}{x - 1} = f'(1) = 17(1)^{16} = 17$$

(ii) $\lim_{h \rightarrow 0} \frac{\sqrt[4]{16+h} - 2}{h}$ (10 pts)

Let $f(x) = \sqrt[4]{x} = x^{1/4}$. The limit is in the form of the derivative definition $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ with $a = 16$ and $f(16) = \sqrt[4]{16} = 2$. The limit is $f'(16)$.

$$f'(x) = \frac{d}{dx} (x^{1/4}) = \frac{1}{4} x^{-3/4} = \frac{1}{4\sqrt[4]{x^3}}$$

$$\lim_{h \rightarrow 0} \frac{\sqrt[4]{16+h} - 2}{h} = f'(16) = \frac{1}{4\sqrt[4]{16^3}} = \frac{1}{4 \cdot 2^3} = \frac{1}{4 \cdot 8} = \frac{1}{32}$$

(iii) $\lim_{\theta \rightarrow \pi/3} \frac{\cos \theta - 0.5}{\theta - \pi/3}$ (10 pts)

Let $f(\theta) = \cos \theta$. The limit is in the form of the derivative definition $f'(a) = \lim_{\theta \rightarrow a} \frac{f(\theta) - f(a)}{\theta - a}$ with $a = \pi/3$ and $f(\pi/3) = \cos(\pi/3) = 1/2 = 0.5$. The limit is $f'(\pi/3)$.

$$f'(\theta) = \frac{d}{dx} (\cos \theta) = -\sin \theta$$

$$\lim_{\theta \rightarrow \pi/3} \frac{\cos \theta - 0.5}{\theta - \pi/3} = f'(\pi/3) = -\sin(\pi/3) = -\frac{\sqrt{3}}{2}$$

Exercise 2.9 (10 pts)

Suppose f is a differentiable function such that $f(g(x)) = x$ and $f'(x) = 1 + [f(x)]^2$. Show that $g'(x) = 1/(1 + x^2)$. Given $f(g(x)) = x$. Differentiate both sides with respect to x using the Chain Rule:

$$f'(g(x))g'(x) = 1$$

We are also given $f'(x) = 1 + [f(x)]^2$. Substitute $g(x)$ for x :

$$f'(g(x)) = 1 + [f(g(x))]^2$$

Since $f(g(x)) = x$, we have:

$$f'(g(x)) = 1 + x^2$$

Substitute this expression for $f'(g(x))$ into the differentiated equation:

$$(1 + x^2)g'(x) = 1$$

Therefore,

$$g'(x) = \frac{1}{1 + x^2}$$

Exercise 2.10 (10 pts)

Show that the length of the portion of any tangent line to the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ cut off by the coordinate axes is constant.

Let $x^{2/3} + y^{2/3} = a^{2/3}$. Differentiate implicitly with respect to x :

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x^{-1/3}}{y^{-1/3}} = -\left(\frac{y}{x}\right)^{1/3}$$

Let (x_0, y_0) be a point on the astroid. The slope of the tangent line at this point is $m = -\left(\frac{y_0}{x_0}\right)^{1/3}$. The equation of the tangent line is:

$$y - y_0 = m(x - x_0) \implies y - y_0 = -\left(\frac{y_0}{x_0}\right)^{1/3} (x - x_0)$$

To find the x -intercept, set $y = 0$:

$$-y_0 = -\left(\frac{y_0}{x_0}\right)^{1/3} (x - x_0) \implies y_0^{2/3} = x_0^{-1/3} (x - x_0)$$

$$x - x_0 = x_0^{1/3} y_0^{2/3} \implies x_{\text{int}} = x_0 + x_0^{1/3} y_0^{2/3}$$

Using $y_0^{2/3} = a^{2/3} - x_0^{2/3}$ (from the astroid equation):

$$x_{\text{int}} = x_0 + x_0^{1/3} (a^{2/3} - x_0^{2/3}) = x_0 + a^{2/3} x_0^{1/3} - x_0 = a^{2/3} x_0^{1/3}$$

To find the y -intercept, set $x = 0$:

$$y - y_0 = -\left(\frac{y_0}{x_0}\right)^{1/3} (0 - x_0) = \left(\frac{y_0}{x_0}\right)^{1/3} x_0$$

$$y_{\text{int}} = y_0 + x_0^{2/3} y_0^{1/3}$$

Using $x_0^{2/3} = a^{2/3} - y_0^{2/3}$:

$$y_{\text{int}} = y_0 + (a^{2/3} - y_0^{2/3}) y_0^{1/3} = y_0 + a^{2/3} y_0^{1/3} - y_0 = a^{2/3} y_0^{1/3}$$

The length L of the segment of the tangent line cut off by the axes is the distance between the intercepts $(x_{\text{int}}, 0)$ and $(0, y_{\text{int}})$:

$$L^2 = (x_{\text{int}} - 0)^2 + (0 - y_{\text{int}})^2 = (a^{2/3} x_0^{1/3})^2 + (a^{2/3} y_0^{1/3})^2$$

$$L^2 = a^{4/3} x_0^{2/3} + a^{4/3} y_0^{2/3} = a^{4/3} (x_0^{2/3} + y_0^{2/3})$$

Since (x_0, y_0) is on the astroid, $x_0^{2/3} + y_0^{2/3} = a^{2/3}$:

$$L^2 = a^{4/3} (a^{2/3}) = a^{4/3+2/3} = a^{6/3} = a^2$$

$$L = \sqrt{a^2} = a$$

Since a is a constant, the length L is constant.

Exercise 2.11 (10 pts)

If $y = \frac{x}{\sqrt{a^2-1}} - \frac{2}{\sqrt{a^2-1}} \arctan\left(\frac{\sin x}{a+\sqrt{a^2-1}+\cos x}\right)$, show that $y' = \frac{1}{a+\cos x}$.

Let $b = \sqrt{a^2-1}$.

$$y = \frac{x}{b} - \frac{2}{b} \arctan\left(\frac{\sin x}{a+b+\cos x}\right)$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{x}{b}\right) - \frac{d}{dx}\left(\frac{2}{b} \arctan\left(\frac{\sin x}{a+b+\cos x}\right)\right)$$

$$\frac{dy}{dx} = \frac{1}{b} - \frac{2}{b} \cdot \frac{1}{1 + \left(\frac{\sin x}{a+b+\cos x}\right)^2} \cdot \frac{d}{dx}\left(\frac{\sin x}{a+b+\cos x}\right)$$

Let $u = \frac{\sin x}{a+b+\cos x}$.

$$\frac{d}{dx}(u) = \frac{\cos x(a+b+\cos x) - \sin x(-\sin x)}{(a+b+\cos x)^2} = \frac{a \cos x + b \cos x + \cos^2 x + \sin^2 x}{(a+b+\cos x)^2}$$

$$= \frac{a \cos x + b \cos x + 1}{(a+b+\cos x)^2}$$

$$1 + u^2 = 1 + \frac{\sin^2 x}{(a+b+\cos x)^2} = \frac{(a+b+\cos x)^2 + \sin^2 x}{(a+b+\cos x)^2}$$

Denominator of $1 + u^2$:

$$(a+b+\cos x)^2 + \sin^2 x = (a+b)^2 + 2(a+b)\cos x + \cos^2 x + \sin^2 x$$

$$= a^2 + 2ab + b^2 + 2(a+b)\cos x + 1$$

$$\text{Since } b^2 = a^2 - 1 \implies b^2 + 1 = a^2 :$$

$$= a^2 + 2ab + a^2 + 2(a+b)\cos x$$

$$= 2a^2 + 2ab + 2(a+b)\cos x$$

$$= 2a(a+b) + 2(a+b)\cos x$$

$$= 2(a+b)(a+\cos x)$$

So,

$$\frac{1}{1+u^2} = \frac{(a+b+\cos x)^2}{2(a+b)(a+\cos x)}$$

Now, substitute back into $\frac{dy}{dx}$:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{1}{b} - \frac{2}{b} \cdot \frac{(a+b+\cos x)^2}{2(a+b)(a+\cos x)} \cdot \frac{a\cos x + b\cos x + 1}{(a+b+\cos x)^2} \\
&= \frac{1}{b} - \frac{1}{b(a+b)(a+\cos x)} \cdot (a\cos x + b\cos x + 1) \\
&= \frac{1}{b} - \frac{1}{b(a+b)(a+\cos x)} \cdot (a\cos x + b\cos x + a^2 - b^2) \quad \text{Since } b^2 + 1 = a^2 \\
&= \frac{1}{b} - \frac{1}{b(a+b)(a+\cos x)} \cdot (\cos x(a+b) + (a+b)(a-b)) \\
&= \frac{1}{b} - \frac{(a+b)(\cos x + a - b)}{b(a+b)(a+\cos x)} \\
&= \frac{a + \cos x}{b(a + \cos x)} - \frac{\cos x + a - b}{b(a + \cos x)} \\
&= \frac{1}{a + \cos x}
\end{aligned}$$

(The assumption $a^2 > 1$ is needed for $b = \sqrt{a^2 - 1}$ to be real).